

PROTOCOL REFERENCE

Laser Competition Target

BLE and Serial Communication Protocol

Accepted working standard

This document records the current desktop application's hardware-working communication contract. Preserve these wire formats, routes, terminators, and state rules unless a later firmware change is deliberately verified.

At a glance

Concern	BLE	Serial / USB
Connection	GATT service with Command and Settings characteristics	One serial byte stream
Settings write	Two messages: FU1 then FU2	One FUK-shaped line
Outgoing terminator	None	CRLF (\r\n)
Incoming framing	Split on CR or LF	Split on CR or LF; 200 ms idle flush
Settings read	Shared incoming FUK format	Shared incoming FUK format

Non-negotiable state rule

Every valid incoming FUK must remain visible in the originating transport's terminal and must immediately replace that transport's Settings state. BLE and serial state stay independent.

Scope

Covers transport setup, routing, message framing, settings writes, FUK reads, SIP time sync, INFO discovery, and the desktop application's state behavior. Lane/target number is currently fixed to 01.

Document version 1.0 | 03 September 2026 | Desktop implementation

1 Transport contracts

The same target can be connected over BLE and serial at the same time. Each transport has independent connection, terminal, SIP mode, and Settings state.

BLE identifiers

GATT item	UUID	Purpose
Service	0bd51666-e7cb-469b-8e4d-2742f1ba77cc	Advertised target service
Command characteristic	e7add780-b042-4876-aae1-11285535f821	Raw device commands and normal incoming notifications
Settings characteristic	e7add780-b042-4876-aae1-11285535f721	FU1/FU2 settings, SIP, and INFO writes; also subscribed when available

BLE operation	Route	Framing
Raw target command	Command characteristic	Plain ASCII; no line ending
Settings update	Settings characteristic	FU1 then FU2; no line ending
SIP / INFO	Settings characteristic	Plain ASCII; no line ending
Incoming messages	Usually Command notifications; accept either subscribed characteristic	Buffer fragments; split complete lines on CR or LF

Serial / USB contract

Property	Current contract
Port model	One continuous byte stream through the serialport package
Baud rate	Application default: 115200; UI-selectable alternatives: 9600, 19200, 38400, 57600, 230400
Outgoing framing	Append CRLF to every protocol command; drain after settings writes
Incoming framing	Split on CR or LF. After 200 ms with no new bytes, flush a final unterminated buffered line.
Routing	One port only; there is no characteristic distinction

Framing rule

Do not copy serial CRLF framing to BLE. BLE writes are complete GATT messages. Conversely, do not send unterminated serial commands: the device may continue waiting for the end of the line.

2 Settings write protocol

The settings model is shared, but the wire command is intentionally different for BLE and serial. Settings controls are available only after the selected transport confirms Standby mode (S).

BLE: two GATT writes

```
FU1:01:0<brightness>:00:<mode>:0<shootingArea>  
FU2:0<shotsHeat>:<secondsHeat>:<HH:mm>
```

Example: FU1:01:03:00:0:02
FU2:05:40:14:07

Serial: one CRLF-terminated line

```
FUK:01:0<brightness>:00:<mode>:0<shootingArea>:0<shotsHeat>:<secondsHeat>:0000000\r\n
```

Example: FUK:01:03:00:0:02:05:40:0000000\r\n

Field contract

Field	Allowed value	BLE encoding	Serial encoding
Target / lane	01 (fixed)	FU1 field 1	FUK field 1
Brightness	1-5	01-05	01-05
Battery placeholder	00	FU1: 00	FUK: 00
Mode	0 Competition; 1 Training; 2 OCR	Single digit	Single digit
Shooting area	0-2	00-02	00-02
Shots per heat	1-5	FU2: 01-05	FUK: 01-05
Heat seconds	10, 20, 30, 40, 50	FU2: literal two digits	FUK: literal two digits
Time / timestamp	Transport-specific	Local HH:mm	Fixed 0000000

Write confirmation

If a requested value already equals the latest device state, no write is sent. Otherwise the app waits up to 10 seconds for the next valid FUK on that transport and resolves with the device-reported values. Timeout or disconnect is an error.

3 FUK read protocol and live state

BLE and serial use the same incoming FUK structure and parser. Only transport delivery and framing differ.

FUK:<targetNumber>:<brightness>:<battery>:<mode>:<shootingArea>:<shotsHeat>:<secondsHeat>:<timestamp>
Example: FUK:01:03:00:0:02:05:40:1234567

Index	Wire field	Stored in Settings?	Meaning
0	FUK	No	Required case-sensitive prefix
1	targetNumber	No	Target / lane identifier
2	brightness	Yes	Brightness selection
3	battery	No	Battery value; remains in raw terminal text
4	mode	Yes	0 Competition, 1 Training, 2 OCR
5	shootingArea	Yes	Selected shooting area
6	shotsHeat	Yes	Maximum shots per heat
7	secondsHeat	Yes	Heat time limit in seconds
8	timestamp	No	Device timestamp; remains in raw terminal text

Every valid FUK follows this path

1	Frame	BLE/serial bytes become one complete text line.
2	Terminal	Publish the raw incoming line to the originating transport's terminal.
3	Parse	Read brightness, mode, shooting area, shots, and seconds.
4	Main state	Replace that transport service's currentSettings.
5	Renderer state	Push device:settings-changed; update only matching BLE or serial Settings.
6	Confirmation	Also resolve a pending settings write, if one exists for that transport.

Parser boundary

The canonical FUK includes timestamp. The app requires fields through secondsHeat and finite numeric values for the five stored settings. It ignores target, battery, and timestamp, tolerates an absent or extra timestamp, and does not range-check numeric values.

There is no dedicated settings-read request on connect. Defaults remain visible until the first valid FUK arrives. FUK has no correlation ID, so the next valid FUK on a transport also confirms any pending write on that transport.

4 Shared commands and replies

SIP and INFO use the same command text on both transports; only their routing and line termination differ.

SIP time synchronization

```
SIP:<lane>:<S|L>:hhmmsscc
Example: SIP:01:L:08263283
```

- lane is fixed to 01.
- S or L is the selected sync kind; the app treats the echoed kind as the confirmed Standby/Live state.
- hhmmsscc is local hours, minutes, seconds, and centiseconds: eight zero-padded digits.
- Observed reply example: SIP:01:L:08263283:11714. The extra final field is device-defined and currently not interpreted.

INFO master discovery

```
INFO<lane>
Example request: INFO01
Example replies: IN0:95:88:91:70
                 IN1:100:93:-1:84
```

- IN<rowIndex>:<value1>:<value2>:... carries one discovery row; documented row indexes are 0-9.
- Invalid/non-numeric values normalize to -1 in the parser. The desktop currently recognizes and highlights rows but does not store a discovery table.

Command	BLE	Serial	Reply handling
SIP	Settings characteristic; no terminator	Single port; append CRLF	Renderer reads echoed S/L kind
INFO	Settings characteristic; no terminator	Single port; append CRLF	IN rows recognized/highlighted
Raw target command	Command characteristic; no terminator	Single port; caller-provided bytes	Terminal receives device output

Recognized incoming message families

Prefix	Desktop behavior
FUK	Terminal + parsed Settings state + pending-write confirmation
SIP	Terminal + confirmed S/L UI mode
IN<number>	Terminal recognition/highlighting; parser available
TAP	Known incoming shot line; terminal display
HCP	Known incoming heat summary line; terminal display

Implementation checklist

Transport and framing	State and confirmation
1. Keep BLE and serial connections and terminals independent.	4. Send every FUK to the terminal and originating transport's Settings state.
2. Use exact BLE characteristic routing; never add serial endings to BLE.	5. Gate settings on confirmed Standby and trust device-reported FUK values.
3. Append CRLF to serial commands; split incoming data on CR or LF.	6. Treat timeout or disconnect as unconfirmed; never apply only the requested value.